Advanced Programming Project

June 2026

Applied Mechatronics

Introduction

Notes on completing the assignment:

Please read the assignment carefully and focus on the topic (understanding the assignment). The quality of the literature review and formal aspects (e.g., spelling and appropriate use of graphics) will be included in the assessment of the paper.

Use of AI tools
If you use AI tools to create source code, you must indicate this.
Include your most relevant prompts including the AI outputs in the appendix to your work, along with the steps you took to correct any errors found in the AI-generated code. Discuss the benefits of the AI tool used.

Implementation and documents to be submitted
This task involves the technical implementation of a manufacturing production line. You are free to select the product to be manufactured, as long as the production line has a minimum of 4 stages or components (Recommended products: Ink marker (tip, ink container, body and cap) or pencil (graphite core, body, eraser and eraser holder))

Final Project Submission
To submit the final project, send an email with the subject StudentID_Lastname_AP, that contains the following:

1. A PDF document explaining how your project was implemented (Use the document Project_Template, the title of the document should be "StudentID_Lastname_AP")
2. A link to your website where the project is hosted and described. (Example: https://advancedprogrammingsrh.github.io/)

Program development

100 points, programming code plus approx. 7-10 pages of accompanying documentation)

This task involves the technical implementation of a manufacturing production line. The project will be evaluated according to the following points.

Task 1 (10 points)
Introduction: Describe the product to be assembled and the production line concept (at least 4 steps or components). Recommended products:

- Ink marker (tip, ink container, body and cap) or
- pencil (graphite core, body, eraser and eraser holder)

Task 2 (60 points)
Program the production line: Create a program to simulate the production line meant to assemble the product you have chosen. It should contain at least:

- Backend – Logic of the program coded in python. It should clearly indicate when a product is marked as defective and why.
- Frontend – HMI (Add the required buttons to operate the machine, such as start, stop and reset). Show the production state and display errors if the machine is faulted.
- Database – InfluxDB
- Dashboard – Grafana (Send at least one parameter to Grafana and display it. In class we used Temperature, State of the machine and parts produced)

Optional modules you could add are:

- Quality Control
- CMMS
- SCADA

Task 3 (20 points)
Explain how you used different tools such as Github, git (Version control), Docker, AI tools such as copilot, Chat GPT, Gemini, etc. Add a link to your website and briefly explain what your project is about (1-2 images and max 4 paragraphs).

Task 4 (10 points)
Conclusion: Discuss your solution. What are the weaknesses of your implementation? What do you like about it? Where could further development be needed? Consider the operation of such a solution. What requirements come into play here?